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A Technical Lemma

For completeness, we include the concentration results from (Mardia et al., 2020) which we used to prove Theorem 3. For a probability distribution $\mathbf{p}$ on $[s]$ define

$$\pi_{\mathbf{p}} := \max_{A \subseteq [s]} \min\{\mathbf{p}(A), 1 - \mathbf{p}(A)\} \quad (12)$$

and notice that $\pi_{\mathbf{p}} \leq \frac{1}{2}$. Define the function $\varphi : [0, 1/2] \rightarrow \mathbb{R}$ by

$$\varphi(x) := \frac{1}{1-2x} \log \frac{1-x}{x}$$

and extend φ by continuity to $\varphi(1/2) := 2$. Observe that φ is strictly decreasing and $2 \leq \varphi(x) < \infty$. The following Lemma concerns the concentration of empirical distribution which captures the dependence on the sample size n , dimension s as well as the structure of the underlying probability distribution via $\varphi(\pi_{\mathbf{p}})$.

Lemma 1. *Let p be a probability distribution on $[s]$ and $\hat{p}$ be the associated empirical distribution obtained from n samples. Then for $\epsilon > 0$ we have*

$$\mathbb{P}(\|\hat{\mathbf{p}} - \mathbf{p}\|_1 \geq \epsilon) \leq \exp\left(-\frac{n\varphi(\pi_{\mathbf{p}})\epsilon^2}{4}\right) G_n(s),$$

where $C_0 = \frac{e^3}{2\pi}$, $C_1 = \frac{6e}{\pi^{3/2}}$ and

$$G_n(s) = \begin{cases} C_1 s (C_0 n/s)^{s/2} & \text{if } \frac{C_0 n}{4} + 2 \leq s \leq \frac{C_0 n}{e} + 2; \\ C_1 s e^{\frac{C_0 n}{2e}} & \text{if } \frac{C_0 n}{e} + 2 \leq s; \\ (2^s - 2) & \text{if } s \leq \frac{C_0 n}{4} + 2. \end{cases} \quad (13)$$

The upper bound

$$\mathbb{P}(\|\hat{\mathbf{p}} - \mathbf{p}\|_1) \leq (2^s - 2) \exp\left(-\frac{n\varphi(\pi_{\mathbf{p}})\epsilon^2}{4}\right)$$

in fact holds for all values of s and n (Weissman et al., 2003), but this bound is very poor when s/n is large, which is commonly the case for language models. Lemma 1 provides an improvement in the regime of small sample size $n \lesssim s$.

B General Results and Proofs

B.1 Derivation of Formula (2)

Proof. The categorical cross-entropy loss reads as

$$\mathcal{L}(\mathbf{w}_1, \dots, \mathbf{w}_s) = -\frac{1}{M} \sum_{i=1}^M \sum_{k=1}^s y_{ik} \log \left(\frac{\exp(\mathbf{w}_k^\top \mathbf{x}_i)}{\sum_{j=1}^K \exp(\mathbf{w}_j^\top \mathbf{x}_i)} \right).$$

The gradient of the loss is expressed as

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}_k} = \frac{1}{M} \sum_{i=1}^M \sum_{\ell=1}^s y_{i\ell} \frac{\frac{\partial}{\partial \mathbf{w}_k} \left[\sum_{j \neq \ell} \exp((\mathbf{w}_j - \mathbf{w}_\ell)^\top \mathbf{x}_i) \right]}{1 + \sum_{j \neq \ell} \exp((\mathbf{w}_j - \mathbf{w}_\ell)^\top \mathbf{x}_i)},$$

where

$$\frac{\partial}{\partial \mathbf{w}_k} \left[\sum_{j \neq \ell} \exp((\mathbf{w}_j - \mathbf{w}_\ell)^\top \mathbf{x}_i) \right] = \begin{cases} -\sum_{j \neq k} \exp((\mathbf{w}_j - \mathbf{w}_k)^\top \mathbf{x}_i) \mathbf{x}_i & \text{if } k = \ell, \\ \exp((\mathbf{w}_k - \mathbf{w}_\ell)^\top \mathbf{x}_i) & \text{if } k \neq \ell. \end{cases}$$

Therefore,

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \mathbf{w}_k} &= \frac{1}{M} \sum_{i=1}^M \left\{ y_{ik} \frac{-\sum_{j \neq k} \exp((\mathbf{w}_j - \mathbf{w}_k)^\top \mathbf{x}_i)}{1 + \sum_{j \neq k} \exp((\mathbf{w}_j - \mathbf{w}_k)^\top \mathbf{x}_i)} + \sum_{\ell \neq k} y_{i\ell} \frac{\exp((\mathbf{w}_k - \mathbf{w}_\ell)^\top \mathbf{x}_i)}{1 + \sum_{j \neq \ell} \exp((\mathbf{w}_j - \mathbf{w}_\ell)^\top \mathbf{x}_i)} \right\} \mathbf{x}_i \\ &= \frac{1}{M} \sum_{i=1}^M \left\{ \sum_{\ell \neq k} y_{i\ell} \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_i)}{\sum_{j=1}^s \exp(\mathbf{w}_j^\top \mathbf{x}_i)} - y_{ik} \frac{\sum_{j \neq k} \exp(\mathbf{w}_j^\top \mathbf{x}_i)}{\sum_{j=1}^s \exp(\mathbf{w}_j^\top \mathbf{x}_i)} \right\} \mathbf{x}_i \\ &= \frac{1}{M} \sum_{i=1}^M \left\{ \sum_{\ell \neq k} y_{i\ell} \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_i)}{\sum_{j=1}^s \exp(\mathbf{w}_j^\top \mathbf{x}_i)} - y_{ik} \left(1 - \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_i)}{\sum_{j=1}^s \exp(\mathbf{w}_j^\top \mathbf{x}_i)} \right) \right\} \mathbf{x}_i \\ &= \frac{1}{M} \sum_{i=1}^M \left\{ \underbrace{\sum_{\ell=1}^s y_{i\ell} \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_i)}{\sum_{j=1}^s \exp(\mathbf{w}_j^\top \mathbf{x}_i)}}_{=1} - y_{ik} \right\} \mathbf{x}_i \\ &= \frac{1}{M} \sum_{i=1}^M \left\{ \frac{\exp(\mathbf{w}_k^\top \mathbf{x}_i)}{\sum_{j=1}^s \exp(\mathbf{w}_j^\top \mathbf{x}_i)} - y_{ik} \right\} \mathbf{x}_i \end{aligned}$$

Finally, solving for $\frac{\partial \mathcal{L}}{\partial \mathbf{w}_k} = \mathbf{0}$ yields (2). $\square$

B.2 Proofs for the Fully Synthetic Case

Proof of Theorem 1. Write $p_i^{(m)} = X_i^{(m)} / n$ where $X_i^{(m)} \sim B(n, p_i^{(m-1)})$ is binomial when conditioned on $\mathbf{p}^{(m-1)}$. So we have

$$\begin{aligned} \mathbb{E}[p_i^{(m)2} \mid \mathbf{p}^{(m-1)}] &= \frac{1}{n} p_i^{(m-1)} + \left(1 - \frac{1}{n}\right) p_i^{(m-1)2} \\ \sum_{i=1}^s \mathbb{E}[p_i^{(m)2} \mid \mathbf{p}^{(m-1)}] &= \frac{1}{n} + \left(1 - \frac{1}{n}\right) \sum_{i=1}^s p_i^{(m-1)2} \end{aligned}$$

and by the law of total expectation

$$S_m = \frac{1}{n} + \left(1 - \frac{1}{n}\right) S_{m-1} = 1 - \left(1 - \frac{1}{n}\right)^m (1 - S_0). \quad (14)$$

Note that $S_m \nearrow 1$ as $m \rightarrow \infty$ and

$$S_m = \mathbb{E} \left[\sum_i p_i^{(m)} p_i^{(m)} \right] \leq \mathbb{E} \left[\left(\sum_i p_i^{(m)} \right) \|\mathbf{p}^{(m)}\|_\infty \right] = \mathbb{E} \|\mathbf{p}^{(m)}\|_\infty.$$

Recall that $\rho_m := \mathbb{P}(\|\mathbf{p}^{(m)}\|_\infty = 1)$ and $T := \inf\{m \in \mathbb{N} : \|\mathbf{p}^{(m)}\|_\infty = 1\} \geq 1$. Since $\sigma_m \geq 1/\tilde{s}$ and $\|\mathbf{p}^{(m)}\|_\infty \in \{\frac{1}{n}, \dots, \frac{n-1}{n}, 1\}$, we have

$$1 \cdot \rho_m + \frac{1}{\tilde{s}}(1 - \rho_m) \leq \mathbb{E} \sigma_m = S_m \leq \mathbb{E} \|\mathbf{p}^{(m)}\|_\infty \leq 1 \cdot \rho_m + (1 - 1/n)(1 - \rho_m),$$

from which (7) follows thanks to (14).

For $k = 1, 2, \dots$ we have $\mathbb{P}(T > k) = \mathbb{P}(\|\mathbf{p}^{(k)}\|_\infty < 1) = 1 - \rho_k$ and thus

$$\frac{1 - S_0}{1 - 1/\bar{s}}(1 - 1/n)^k \leq \mathbb{P}(T > k) \leq n(1 - S_0)(1 - 1/n)^k,$$

which establishes (8) because $\mathbb{E}[T] = \sum_{k=0}^\infty \mathbb{P}(T > k)$. $\square$

Proof of Proposition 1. Fix $i \in [s]$. For $m \geq 1$ consider the events $E_m := \{T \leq m\}$ and $F_m := \{\mathbf{p}_i^{(m)} = 1\}$. Then $\mathbf{p}_i^{(m)} \in \{0, 1\}$ on E_m and $F_m \subseteq E_m$. Observe that

$$E_m \nearrow \cup_m E_m \quad \text{and} \quad F_m \nearrow \cup_m F_m = \{\lim_{m \rightarrow \infty} \mathbf{p}^{(m)} = \delta_i\},$$

where $\mathbb{P}(\cup_m E_m) = 1$ by Theorem 1 and in particular $\mathbb{P}(E_m) > 0$ for m large. Thus

$$\begin{aligned} p_i &= \lim_{m \rightarrow \infty} \mathbb{E}[p_i^{(m)}] = \lim_{m \rightarrow \infty} [\mathbb{P}(E_m) \mathbb{E}[p_i^{(m)} | E_m] + \mathbb{P}(E_m^c) \mathbb{E}[p_i^{(m)} | E_m^c]] \\ &= \lim_{m \rightarrow \infty} \mathbb{E}[p_i^{(m)} | E_m] = \lim_{m \rightarrow \infty} \mathbb{P}(p_i^{(m)} = 1 | E_m) = \lim_{m \rightarrow \infty} \frac{\mathbb{P}(F_m \cap E_m)}{\mathbb{P}(E_m)} \\ &= \lim_{m \rightarrow \infty} \frac{\mathbb{P}(F_m)}{\mathbb{P}(E_m)} = \lim_{m \rightarrow \infty} \mathbb{P}(F_m) = \mathbb{P}(\cup_m F_m), \end{aligned}$$

which proves the assertion. $\square$

B.3 Proofs for the Partially Synthetic Case

Proof of Theorem 2. Let $v_i^{(m)} := \mathbb{E}[p_i^{(m)^2}]$, $N' := N + n$ and $\mathbf{y}_i^{(m)} = (y_{1,i}^{(m)}, \dots, y_{s,i}^{(m)})$. Then

$$v_i^{(1)} = \frac{p_i}{N} + \left(1 - \frac{1}{N}\right) p_i^2$$

and for $m \geq 2$

$$\begin{aligned} v_i^{(m)} &= \frac{1}{N'^2} \mathbb{E} \left[\left(\sum_{k=1}^N y_{i,k}^{(0)} \right)^2 + \left(\sum_{k=1}^n y_{i,k}^{(m-1)} \right)^2 + 2 \sum_{j=1}^N \sum_{k=1}^n y_{i,j}^{(0)} y_{i,k}^{(m-1)} \right] \\ &= \frac{1}{N'^2} \left[N p_i + (N^2 - N) p_i^2 + n p_i + (n^2 - n) v_i^{(m-1)} \right] + \frac{2}{N'^2} \sum_{j=1}^N \sum_{k=1}^n \mathbb{E} \left[y_{i,j}^{(0)} y_{i,k}^{(m-1)} \right] \\ &= \frac{p_i}{N'} + \frac{N^2 - N}{N'^2} p_i^2 + \frac{n^2 - n}{N'^2} v_i^{(m-1)} + \frac{2}{N'^2} \sum_{j=1}^N \sum_{k=1}^n \mathbb{E} \left[y_{i,j}^{(0)} y_{i,k}^{(m-1)} \right]. \end{aligned} \tag{15}$$

For $m \geq 2$, since $y_{i,k}^{(m-1)}$ is conditionally independent of $y_{i,j}^{(0)}$ given $p_i^{(m-1)}$, by conditioning on $y_{i,j}^{(0)}$ and $p_i^{(m-1)}$ we have

$$\mathbb{E} \left[y_{i,j}^{(0)} y_{i,k}^{(m-1)} \right] = \mathbb{E} \left[y_{i,j}^{(0)} p_i^{(m-1)} \right]$$

and hence for $m \geq 2$,

$$v_i^{(m)} = \frac{p_i}{N'} + \frac{N^2 - N}{N'^2} p_i^2 + \frac{n^2 - n}{N'^2} v_i^{(m-1)} + \frac{2n}{N'^2} \sum_{j=1}^N \mathbb{E} \left[y_{i,j}^{(0)} p_i^{(m-1)} \right].$$